

Problem 23.1

$$\frac{d\|u(t)\|^2}{dt} = 2u_1 u_1' + 2u_2 u_2' + 2u_3 u_3' = 0$$

$u(t)$ never changes length.

Problem 23.2

$$\begin{aligned} S e^{At} S^{-1} &= \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix} \\ &= \begin{bmatrix} e^t & .5e^{3t} - .5e^t \\ 0 & e^{3t} \end{bmatrix} \end{aligned}$$